

Computer vision - Assignment 3

1. Download the image “moon.jpg”. Apply the following spatial filters to the image and compare the results by plotting the images side-by-side:
 - (a) Box filter with kernel size 9×9
 - (b) Gaussian filter
 - (c) Laplacian filter - apply the filter and display the Laplacian. Then use appropriate image addition to enhance the edges. Display the result.
 - (d) Sobel filter

What are your observations about the effects of each of these filters on the given image?

2. On the Viola-Jones algorithm (refer to the paper):
 - (a) State the main steps and features of the Viola-Jones algorithm.
 - (b) Consider an image of size 6×6 filled with integers between 0 and 7 as follows:

0	0	2	3	0	0
0	2	3	3	2	0
0	0	5	7	0	0
0	0	2	4	0	0
1	2	0	0	1	0
1	0	0	0	1	0

Make a list (and give a count) of all the rectangle features that will be calculated by the VJ algorithm for this image.

- (c) For the image in part (ii), compute the integral image and explain the comment below Figure 3 of the Viola-Jones paper.
 - (d) Use the above part to calculate the values of the rectangle features. It is recommended that you program a suitable code for this calculation.
3. Consider the following 8×8 grayscale image patch representing a single HOG cell:

$$I(x, y) = 3x + y, \quad x, y = 0, 1, \dots, 7.$$

Assume:

- Simple centered finite differences:

$$G_x = I(x + 1, y) - I(x - 1, y), \quad G_y = I(x, y + 1) - I(x, y - 1).$$

- Unsigned gradients (0–180°).
- 9 orientation bins.

Compute the final 9-dimensional histogram for the entire cell by following these steps for each cell:

- (a) Compute (G_x, G_y) at any interior pixel.
- (b) Compute gradient magnitude and orientation.
- (c) Determine which two bins receive votes using the interpolation method explained at the end of the assignment.

4. Consider an input RGB image of size

$$32 \times 32 \times 3.$$

A convolutional neural network is defined as follows:

Layer 1: Convolution

- Number of filters: 16
- Filter size: 3×3
- Padding: 1
- Stride: 1

Layer 2: Max Pooling

- Window size: 2×2
- Stride: 2
- No padding

Layer 3: Convolution

- Number of filters: 32
- Filter size: 3×3
- Padding: 1
- Stride: 1

Layer 4: Max Pooling

- Window size: 2×2
- Stride: 2
- No padding

Layer 5: Convolution

- Number of filters: 64
- Filter size: 3×3
- Padding: 1
- Stride: 1

Layer 6: Fully Connected

- 128 hidden units

Layer 7: Fully Connected

- 10 output units

Questions

- (a) Compute the spatial dimensions after each layer.
- (b) What is the dimension of the flattened feature vector before the first fully connected layer?
- (c) How many learnable parameters (including biases) are there in:
 - i. The first convolutional layer?
 - ii. The third convolutional layer?
 - iii. The first fully connected layer?

Formal Definition of Orientation Interpolation Let a cell contain N pixels indexed by $p = 1, \dots, N$. For each pixel, let

$$m_p = \|\nabla I(p)\| \quad \text{and} \quad \theta_p \in [0, 180^\circ)$$

denote its gradient magnitude and unsigned orientation.

Let the bin centers be

$$\theta_k = k\Delta, \quad k = 0, 1, \dots, 8,$$

with bin width $\Delta = 20^\circ$.

Define the triangular interpolation weight function

$$w_k(\theta) = \max\left(0, 1 - \frac{|\theta - \theta_k|}{\Delta}\right).$$

Then the k -th histogram entry is defined as

$$h_k = \sum_{p=1}^N m_p w_k(\theta_p).$$

Note that for any fixed θ_p , the weights satisfy

$$\sum_k w_k(\theta_p) = 1,$$

so the total gradient energy m_p is preserved under interpolation.

Specifically, the vote of magnitude $m = \|\nabla I\|$ is distributed as:

$$h_k = h_k + m \cdot \frac{\theta_{k+1} - \theta}{\Delta},$$

$$h_{k+1} = h_{k+1} + m \cdot \frac{\theta - \theta_k}{\Delta}.$$

Equivalently, $h_k = h_k + m(1 - \alpha)$, $h_{k+1} = h_{k+1} + m\alpha$, where $\alpha = \frac{\theta - \theta_k}{\Delta}$.